

Distribuciones de Probabilidad

Estephany Carolina Reyes Fuentes

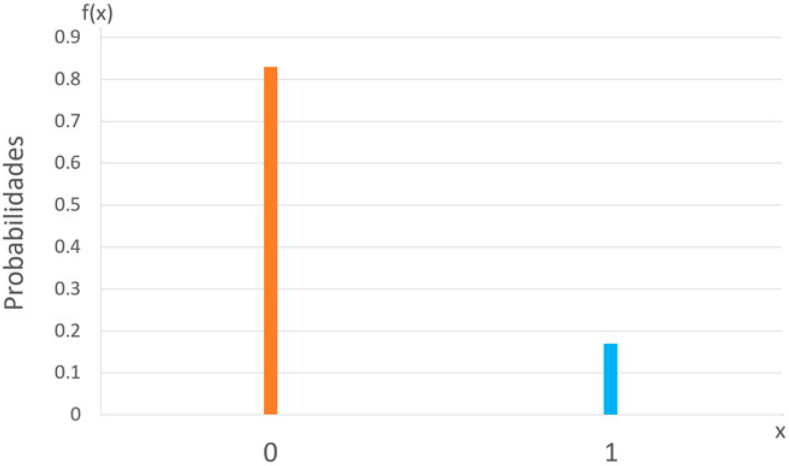
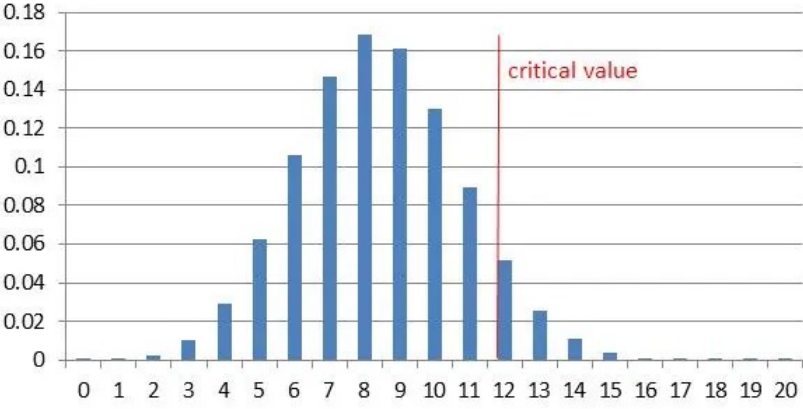
Distribuciones Discretas

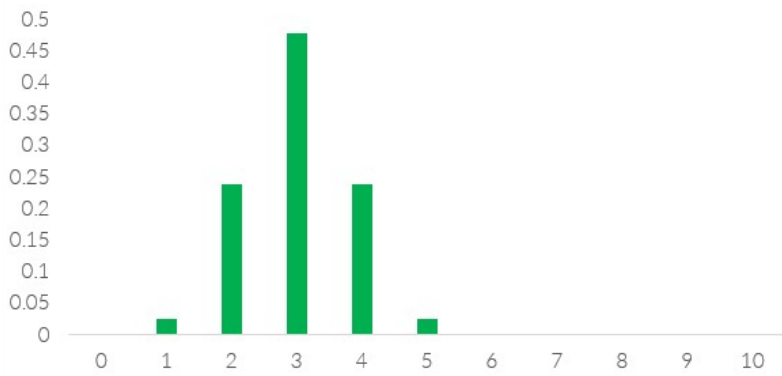
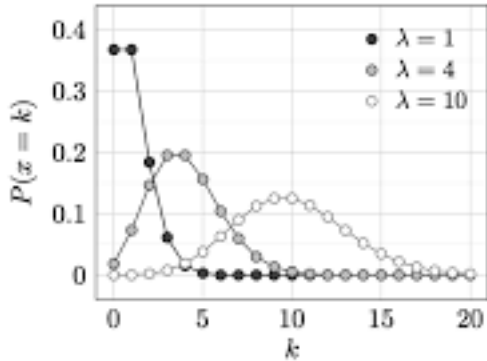
Distribución	Notación	Función de Probabilidad	Función Acumulada	Momentos	FGM
Bernoulli	$X \sim Ber(p)$	$P(X = x) = p^x(1 - p)^{1-x}$	$F(x) = \begin{cases} 0, & x < 0 \\ 1 - p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$	$E(X) = p$ $Var(X) = p(1 - p)$	$M_X(t) = 1 - p + pe^t$
Binomial	$X \sim Bin(n, p)$	$\binom{n}{x} p^x (1 - p)^{n-x}$	$\sum_{k=0}^x \binom{n}{k} p^k (1 - p)^{n-k}$	$E(X) = np$ $Var(X) = np(1 - p)$	$(1 - p + pe^t)^n$
Hipergeométrica	$X \sim H(N, K, n)$	$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$	$F(x) = \sum_{k=0}^x \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$	$E(X) = n \frac{K}{N}$ $Var(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$	$\sum_x e^{tx} P(X = x)$
Poisson	$X \sim Pois(\lambda)$	$\frac{e^{-\lambda} \lambda^x}{x!}$	$\sum_{k=0}^x \frac{e^{-\lambda} \lambda^k}{k!}$	$E(X) = \lambda$ $Var(X) = \lambda$	$e^{\lambda(e^t - 1)}$
Geométrica	$X \sim Geom(p)$	$(1 - p)^{x-1} p, \quad x = 1, 2, \dots$	$1 - (1 - p)^x$	$E(X) = \frac{1}{p}$ $Var(X) = \frac{1-p}{p^2}$	$\frac{pe^t}{1 - (1-p)e^t}$

Binomial Negativa	$X \sim NB(r, p)$	$P(X = x) = \binom{x+r-1}{x} (1-p)^x p^r$	$F(x) = \sum_{k=0}^x \binom{k+r-1}{k} (1-p)^k p^r$	$E(X) = \frac{r(1-p)}{p}$ $Var(X) = \frac{r(1-p)}{p^2}$	$\left(\frac{p}{1 - (1-p)e^t} \right)^r$
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Distribuciones Continuas

Distribución	Notación	Función de Densidad	Función Acumulada	Momentos	FGM
Uniforme	$X \sim U(a, b)$	$\frac{1}{b-a}, \quad a \leq x \leq b$	$\frac{x-a}{b-a}, \quad a \leq x \leq b$	$E(X) = \frac{a+b}{2}$ $Var(X) = \frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}, \quad t \neq 0$
Exponencial	$X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}, \quad x > 0$	$1 - e^{-\lambda x}, \quad x > 0$	$E(X) = \frac{1}{\lambda}$ $Var(X) = \frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}, \quad t < \lambda$
Normal	$X \sim N(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$	$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$	$E(X) = \mu$ $Var(X) = \sigma^2$	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$
Gamma	$X \sim \Gamma(\alpha, \lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0$	$\frac{\gamma(\alpha, \lambda x)}{\Gamma(\alpha)}, \quad x > 0$	$E(X) = \frac{\alpha}{\lambda}$ $Var(X) = \frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda-t} \right)^\alpha, \quad t < \lambda$
Beta	$X \sim Beta(\alpha, \beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$	$I_x(\alpha, \beta)$	$E(X) = \frac{\alpha}{\alpha+\beta}$ $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\int_0^1 e^{tx} f(x) dx$
Chi-cuadrada	$X \sim \chi^2(\nu)$	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}, \quad x > 0$	$\frac{\gamma\left(\frac{\nu}{2}, \frac{x}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)}, \quad x > 0$	$E(X) = \nu$ $Var(X) = 2\nu$	$(1-2t)^{-\nu/2}, \quad t < \frac{1}{2}$

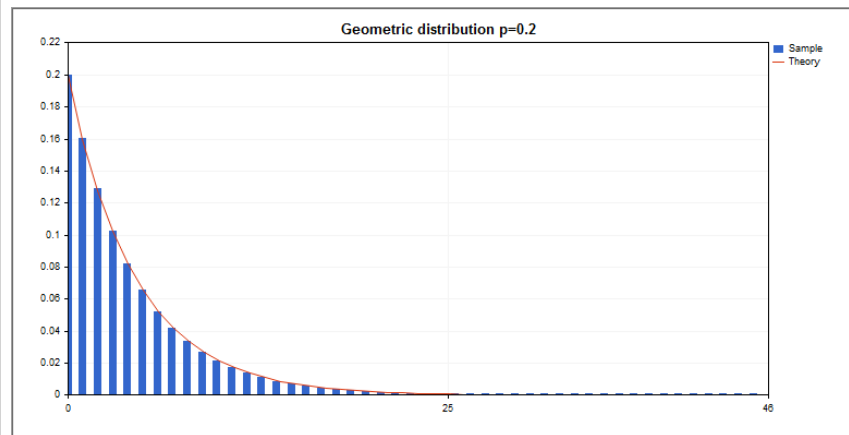
Distribución	Gráfica																																												
Bernoulli	Representación gráfica de la función de probabilidad de X  <table border="1" data-bbox="1087 313 1871 776"> <caption>Data for Bernoulli Distribution Graph</caption> <thead> <tr> <th>x</th> <th>f(x)</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>~0.82</td> </tr> <tr> <td>1</td> <td>~0.18</td> </tr> </tbody> </table>	x	f(x)	0	~0.82	1	~0.18																																						
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Poisson	Distribución de Poisson $f(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \mathbb{N}, \lambda > 0$  <table border="1"><caption>Approximate Data for Poisson Distribution</caption><thead><tr><th>k</th><th>P(k; λ=1)</th><th>P(k; λ=4)</th><th>P(k; λ=10)</th></tr></thead><tbody><tr><td>0</td><td>0.37</td><td>0.02</td><td>0.00</td></tr><tr><td>1</td><td>0.37</td><td>0.07</td><td>0.00</td></tr><tr><td>2</td><td>0.18</td><td>0.14</td><td>0.00</td></tr><tr><td>3</td><td>0.06</td><td>0.21</td><td>0.00</td></tr><tr><td>4</td><td>0.02</td><td>0.21</td><td>0.00</td></tr><tr><td>5</td><td>0.01</td><td>0.17</td><td>0.00</td></tr><tr><td>6</td><td>0.00</td><td>0.12</td><td>0.00</td></tr><tr><td>7</td><td>0.00</td><td>0.07</td><td>0.00</td></tr><tr><td>8</td><td>0.00</td><td>0.04</td><td>0.00</td></tr><tr><td>9</td><td>0.00</td><td>0.02</td><td>0.00</td></tr><tr><td>10</td><td>0.00</td><td>0.01</td><td>0.12</td></tr><tr><td>11</td><td>0.00</td><td>0.00</td><td>0.11</td></tr><tr><td>12</td><td>0.00</td><td>0.00</td><td>0.10</td></tr><tr><td>13</td><td>0.00</td><td>0.00</td><td>0.08</td></tr><tr><td>14</td><td>0.00</td><td>0.00</td><td>0.06</td></tr><tr><td>15</td><td>0.00</td><td>0.00</td><td>0.04</td></tr><tr><td>16</td><td>0.00</td><td>0.00</td><td>0.03</td></tr><tr><td>17</td><td>0.00</td><td>0.00</td><td>0.02</td></tr><tr><td>18</td><td>0.00</td><td>0.00</td><td>0.01</td></tr><tr><td>19</td><td>0.00</td><td>0.00</td><td>0.01</td></tr><tr><td>20</td><td>0.00</td><td>0.00</td><td>0.00</td></tr></tbody></table>	k	P(k; λ=1)	P(k; λ=4)	P(k; λ=10)	0	0.37	0.02	0.00	1	0.37	0.07	0.00	2	0.18	0.14	0.00	3	0.06	0.21	0.00	4	0.02	0.21	0.00	5	0.01	0.17	0.00	6	0.00	0.12	0.00	7	0.00	0.07	0.00	8	0.00	0.04	0.00	9	0.00	0.02	0.00	10	0.00	0.01	0.12	11	0.00	0.00	0.11	12	0.00	0.00	0.10	13	0.00	0.00	0.08	14	0.00	0.00	0.06	15	0.00	0.00	0.04	16	0.00	0.00	0.03	17	0.00	0.00	0.02	18	0.00	0.00	0.01	19	0.00	0.00	0.01	20	0.00	0.00	0.00
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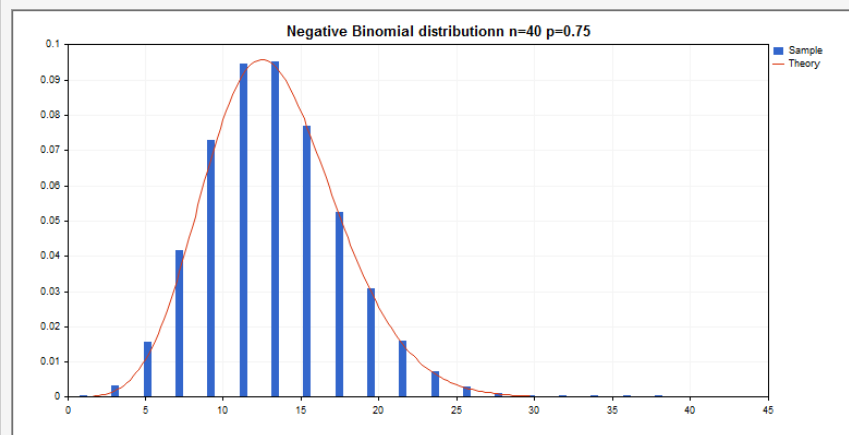
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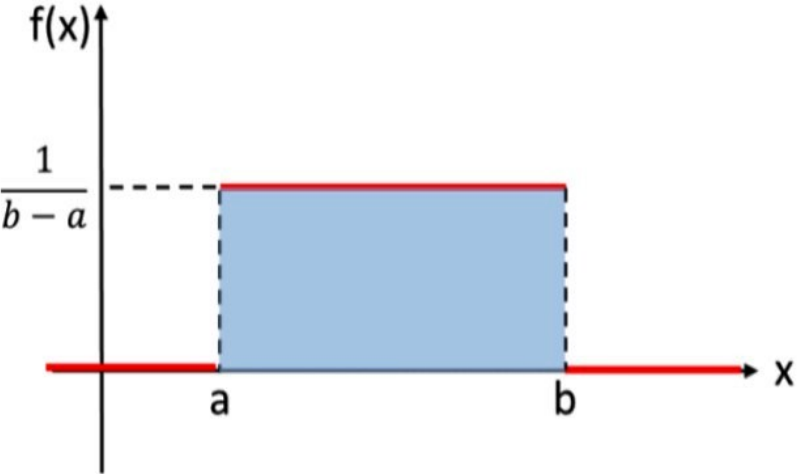
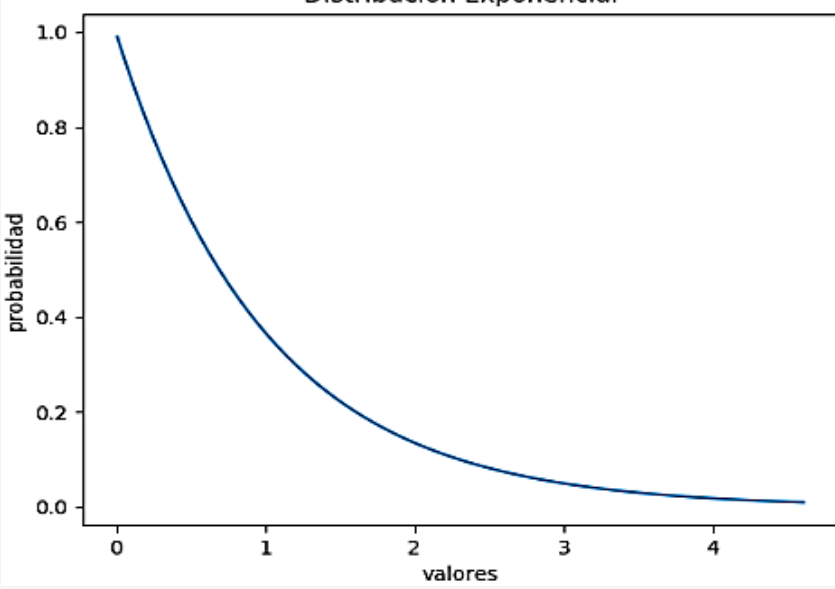
Geométrica

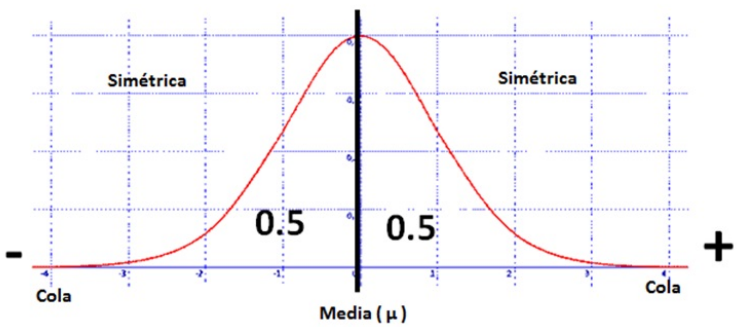
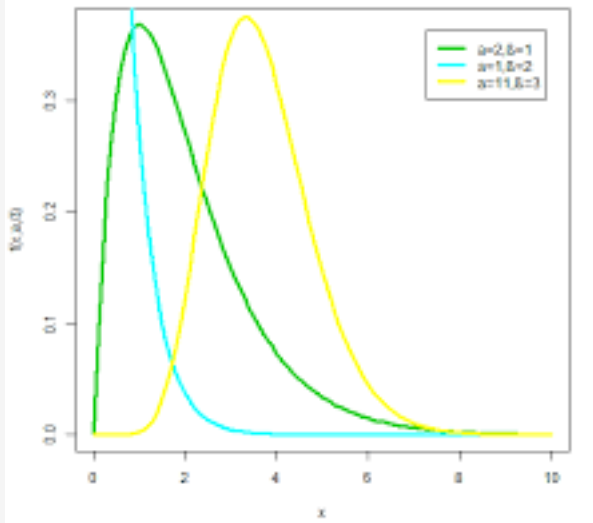
Gráfica

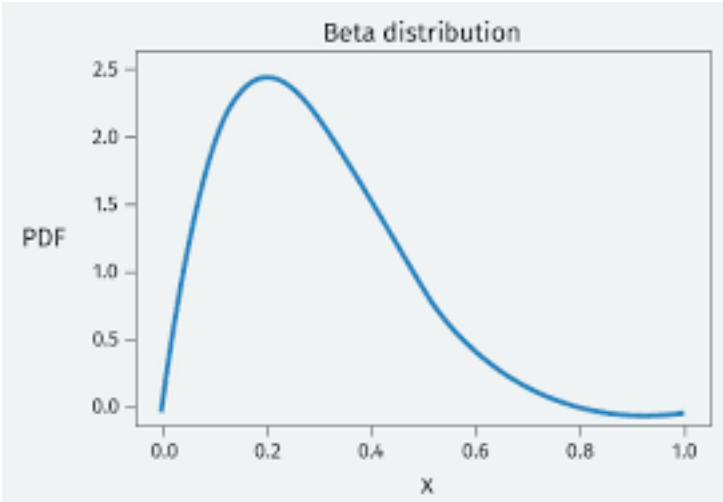


Binomial Negativa



Distribución	Gráfica
Uniforme	 The graph illustrates a Uniform distribution. The horizontal axis is labeled x and the vertical axis is labeled $f(x)$. A blue shaded rectangle represents the probability density function, which is constant over the interval $[a, b]$. The height of the rectangle is labeled $\frac{1}{b-a}$. Dashed lines indicate the boundaries at a and b on the x-axis and the height $\frac{1}{b-a}$ on the $f(x)$-axis.
Exponencial	 The graph illustrates an Exponential distribution. The horizontal axis is labeled "valores" and the vertical axis is labeled "probabilidad". The curve starts at (0, 1.0) and decays towards zero as "valores" increase. The title of the graph is "Distribución Exponencial".

Distribución	Gráfica
Normal	DISTRIBUCIÓN NORMAL  The graph shows a red bell-shaped curve on a grid. A vertical black line marks the center, labeled 'Media (μ)' on the x-axis. The area under the curve is divided into two equal halves by this line, each labeled '0.5'. The word 'Simétrica' (Symmetric) appears on both sides of the center. The x-axis is labeled 'Cola' (Tail) at both ends, with a minus sign on the left and a plus sign on the right.
Gamma	Gráfico distribución Gamma  The graph displays three Gamma distribution curves on a coordinate system. The x-axis is labeled 'x' and ranges from 0 to 10. The y-axis is labeled 'f(x; a, b)' and ranges from 0.0 to 0.3. A legend in the top right corner identifies the curves: a green curve for 'a=2, b=1', a cyan curve for 'a=1, b=2', and a yellow curve for 'a=11, b=3'. The green curve peaks at x=1, the cyan curve peaks at x=0.5, and the yellow curve peaks at x=3.5.

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Beta	 The graph displays the Probability Density Function (PDF) of a Beta distribution. The x-axis, labeled 'x', ranges from 0.0 to 1.0 with major ticks every 0.2. The y-axis, labeled 'PDF', ranges from 0.0 to 2.5 with major ticks every 0.5. The curve is a smooth, blue line that starts at (0,0), rises to a peak of approximately 2.4 at x=0.2, and then gradually declines, approaching zero as x approaches 1.0. The title 'Beta distribution' is centered above the plot area. <table border="1"><caption>Approximate data points for the Beta distribution PDF</caption><thead><tr><th>x</th><th>PDF</th></tr></thead><tbody><tr><td>0.0</td><td>0.00</td></tr><tr><td>0.1</td><td>1.50</td></tr><tr><td>0.2</td><td>2.40</td></tr><tr><td>0.3</td><td>2.10</td></tr><tr><td>0.4</td><td>1.50</td></tr><tr><td>0.5</td><td>0.80</td></tr><tr><td>0.6</td><td>0.40</td></tr><tr><td>0.7</td><td>0.15</td></tr><tr><td>0.8</td><td>0.05</td></tr><tr><td>0.9</td><td>0.01</td></tr><tr><td>1.0</td><td>0.00</td></tr></tbody></table>	x	PDF	0.0	0.00	0.1	1.50	0.2	2.40	0.3	2.10	0.4	1.50	0.5	0.80	0.6	0.40	0.7	0.15	0.8	0.05	0.9	0.01	1.0	0.00
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